

Article

Machine Learning Approaches for Classifying Chess Game Outcomes: A Comparative Analysis of Player Ratings and Game Dynamics

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Abstract

Online chess platforms generate vast amounts of game data, presenting opportunities to analyze match outcomes using machine learning approaches. This study develops and compares four machine learning models to classify chess game results (White win, Black win, or Draw) by integrating player rating information with game dynamic metadata. We analyzed 11,510 complete games from the Lichess platform after preprocessing a dataset of 20,058 initial records. Seven key features were engineered to capture both pre-game skill parameters (player ratings, rating difference) and game complexity metrics (game duration, turn count). Four machine learning algorithms were implemented and optimized through grid search cross-validation: Multinomial Logistic Regression, Random Forest, K-Nearest Neighbors, and Histogram Gradient Boosting. The Gradient Boosting classifier achieved the highest performance with 83.19% accuracy on hold-out data and consistent 5-fold cross-validation scores ($83.08\% \pm 0.009\%$). Feature importance analysis revealed that game complexity (number of turns) was the strongest correlate of the outcome across all models, followed by the rating difference between opponents. Draws represented only 5.11% of outcomes, creating class imbalance challenges that affected classification performance for this outcome category. The results demonstrate that ensemble methods, particularly gradient boosting, can effectively capture non-linear interactions between player skill and game length to classify chess outcomes. These findings have practical applications for chess platforms in automated content curation, post-game quality assessment, and engagement enhancement strategies. The study establishes a foundation for robust outcome analysis systems in online chess environments.

Keywords: chess prediction; machine learning; classification algorithms; online gaming; player rating systems; gradient boosting; game outcome forecasting



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1. Introduction

Chess, one of the world's most popular strategic board games, has experienced a digital renaissance with the emergence of online platforms that host millions of games daily. Platforms such as Lichess, Chess.com, and others generate extensive datasets containing game metadata, player information, and outcome results. This wealth of data presents unique opportunities to apply machine learning techniques for classifying game outcomes, understanding player behavior, and optimizing platform experiences [1].

The prediction of chess game outcomes has significant practical applications. For players, understanding the factors that influence game results can inform training strategies and opening preparation. For platform developers, accurate outcome classification enables automated game tagging, personalized content recommendations, and enhanced user engagement mechanisms. Additionally, such predictive models contribute to the broader understanding of competitive gaming dynamics and skill-based outcome determination.

Previous research in chess outcome prediction has explored various approaches, from traditional statistical methods to advanced machine learning techniques. Early work focused on position evaluation using hand-crafted features, while more recent studies have leveraged deep learning approaches for move prediction and position assessment [2]. However, there remains a gap in systematic comparative analysis of machine learning algorithms for classifying chess outcomes using readily available game metadata.

This study addresses the multiclass classification problem of determining whether an online chess game results in a White win, Black win, or Draw, by integrating pre-game player ratings with post-game complexity metrics. While game duration and turn count are outcome-dependent, they are included here to quantify the relative influence of game struggle (complexity) versus skill disparity (ratings) on the final result.

We compare four distinct machine learning approaches: Multinomial Logistic Regression, Random Forest, K-Nearest Neighbors, and Histogram Gradient Boosting Classifier. These algorithms were selected to represent four distinct families of classifiers—linear, instance-based, bagging ensembles, and boosting ensembles—providing a comprehensive overview of performance across different modeling paradigms.

Our analysis is based on 11,510 complete games from the Lichess platform, providing insights into the relative importance of player skill ratings versus game dynamic factors.

The primary contributions of this work include: (1) a comprehensive comparative analysis of four machine learning algorithms for chess outcome prediction, (2) identification of key outcome determinants from game metadata, (3) quantitative assessment of class imbalance effects on prediction accuracy, and (4) practical recommendations for implementing automated analysis systems in online chess platforms.

2. Literature Review

The application of machine learning to predict outcomes in sports and games has a rich history. In chess, early research focused on rule-based systems and expert systems, but the advent of large datasets and powerful computing has shifted the focus to statistical and machine learning approaches. A significant body of work has explored the use of various machine learning models to predict chess game results, with a strong emphasis on feature engineering and model selection [3].

Several studies have demonstrated the effectiveness of using player ratings, such as the Elo rating system, as a primary feature for outcome prediction. For instance, a study utilized a Hidden Markov Model to predict game outcomes based on player ratings and past performance [4]. Their work highlighted the importance of historical data in capturing the dynamic nature of player skill levels. More recently, researchers have combined Elo ratings with other features, such as opening moves and game complexity metrics, to improve predictive accuracy [5].

The choice of machine learning algorithm has also been a central theme in the literature. While traditional models like logistic regression have been used as a baseline, more advanced ensemble methods, such as Random Forest and Gradient Boosting, have consistently shown superior performance. A study by Lunariz demonstrated the effectiveness of LightGBM, a gradient boosting framework, in predicting chess game results on the Lichess

platform [6]. Their findings align with our own, indicating that the ability of these models to capture complex, non-linear relationships is crucial for high accuracy.

Recent developments in deep learning have opened new avenues for chess analysis. Iavich and Kevanishvili developed a neural network-based approach for detecting fair play violations in chess, achieving approximately 98% accuracy in distinguishing between human and computer-generated play. Their method utilized sequential neural networks analyzing centipawn deviation metrics, time distribution patterns, and insights from Maia neural networks, demonstrating the potential of deep learning for chess game analysis beyond simple outcome prediction [7].

The integration of convolutional neural networks (CNNs) with temporal data has shown promising results. In [8] the authors presented a novel approach combining CNN and bidirectional LSTM architectures to estimate chess player ratings directly from game moves and clock times. Their model achieved a mean absolute error of 182 rating points on test data from over 1.2 million Lichess games, demonstrating the first neural network capable of providing move-by-move rating predictions without hand-crafted features. The inclusion of clock time data proved particularly valuable for faster time controls, improving performance by 24% overall.

The field has also seen significant advances in making chess AI systems more interpretable. Techniques were developed using information-based explanation methods for deep learning chess agents using large open-source models like Leela Chess Zero. Their work demonstrated that concept detection methodologies previously applied to proprietary models like AlphaZero could be successfully replicated on publicly available systems, enabling broader research into how neural networks learn chess concepts and strategies [9].

The IEEE Big Data Cup competitions have provided valuable benchmarks for chess-related machine learning tasks. Zýsko and others reported on the 2024 competition focused on predicting chess puzzle difficulty, which attracted 97 teams making over 1200 submissions. The winning approaches utilized ensemble methods combining deep neural networks (particularly CNNs) with gradient boosting models, achieving mean squared errors below 105,000. This competition highlighted the effectiveness of combining domain-specific feature engineering with modern machine learning architectures [10].

The evaluation of chess positions using neural networks has been refined through systematic investigation of dataset impacts. A study in 2022 conducted experimental research on chess position evaluation using deep neural networks, creating datasets with over 1.6 million unique positions each. Their multilayer perceptron models achieved higher accuracy when trained on carefully curated positions compared to datasets including randomly generated positions, demonstrating the importance of data quality in chess AI development [11].

Focused studies on specific chess scenarios offer complementary insights. A 2024 study analyzed the “king–rook vs. king–pawn” endgame using a dataset from the UCI Machine Learning Repository. After encoding board states and engineering features such as piece placement, king proximity and pawn advancement, they compared logistic regression, decision trees and ensemble classifiers; their results show that careful feature engineering can improve prediction of endgame outcomes and inform strategic recommendations [12].

Recent work has expanded the scope of chess analytics by exploring a variety of datasets, feature sets, and modeling strategies. A graduate thesis compiled 2.5 million games from The Week in Chess and derived six high-level statistics—win and draw rates for White, Black and the opening—to train multiple classifiers. By comparing results across the full and reduced feature sets, she demonstrated that these aggregated statistics contain significant predictive information about game outcomes [13]. Reddy and co-authors approached the problem from a sequential perspective. Their 2023 conference

paper encodes move sequences and employs recurrent neural networks and long short-term memory units to capture temporal dependencies; their dataset includes games with diverse ratings, time controls and openings, and they report promising accuracy on both binary and multi-class outcome prediction tasks with careful hyperparameter tuning [14].

Furthermore, the field of sports analytics has provided a broader context for this research. The application of machine learning in sports has grown rapidly, with a focus on everything from player performance analysis to in-game strategy optimization. As noted in [15], machine learning algorithms are used to analyze match data, track player movements, and gain insights into tactical patterns. This broader trend underscores the potential for data-driven approaches to transform our understanding of competitive games like chess.

Despite these advancements, challenges remain. Class imbalance, particularly the low frequency of draws, is a recurring issue in chess outcome prediction. This can lead to models that are biased towards the majority classes (White or Black wins) and perform poorly on the minority class (draws). Additionally, the selection of features is an ongoing area of research. While pre-game metadata is readily available, incorporating in-game features, such as piece-square tables and engine evaluations, could further enhance predictive power, though at the cost of increased computational complexity.

This study builds upon the existing literature by providing a comprehensive comparison of multiple machine learning models on a curated dataset of online chess games. By focusing on player metadata and game characteristics, we aim to develop an interpretable model that quantifies the determinants of victory. Our work contributes to the growing body of research on machine learning in chess and sports analytics, offering new insights into the factors that drive game outcomes and providing a solid foundation for future research in this exciting field.

3. Methodology

3.1. Dataset Description

The dataset utilized in this study, *games.csv*, comprises 20,058 online chess games downloaded via the Lichess Open Database API [16]. Each record contained comprehensive game metadata including temporal information, player ratings, game settings, and outcome results. Key data fields included:

- Temporal data: *created_at* and *last_move_at* timestamps (Unix milliseconds);
- Player information: *white_rating* and *black_rating* (Elo scores);
- Game metadata: *winner*, *opening_name*, *turns*, *rated* status;
- Additional features: Various game-specific attributes and settings.

Initial data quality assessment revealed that 8548 games (42.5%) had zero-second duration, indicating aborted games, disconnections, or technical issues. These records were excluded from analysis, resulting in a final dataset of 11,510 complete games suitable for outcome prediction modeling.

3.2. Data Preprocessing and Feature Engineering

Data preprocessing involved five key transformation steps to prepare the dataset for machine learning analysis:

1. **Timestamp Conversion:** Unix millisecond timestamps were converted to readable datetime objects for temporal analysis and feature derivation.
2. **Duration Calculation:** Game duration in seconds was computed as the difference between *last_move_at* and *created_at* timestamps.

3. **Rating Difference Engineering:** A new feature `rating_diff` was created by subtracting Black's rating from White's rating (white rating—black rating), capturing the skill differential between opponents in a single numerical value.
4. **Outcome Encoding:** Game results were categorized into three classes: White win, Black win, and Draw. Given the low frequency of draws (5.11%), both binary and multiclass classification approaches were explored. To maintain the integrity of the real-world data distribution and establish a baseline performance, we utilized the natural class distribution without synthetic oversampling (e.g., SMOTE), allowing the models to reflect the genuine rarity of draws in online play.
5. **Categorical Variable Encoding:** The Boolean 'rated' flag was converted to numerical format (0/1) for compatibility with machine learning algorithms.

The final feature set comprised seven variables: `white_rating`, `black_rating`, `rating_diff`, `game_duration_sec`, `turns`, `opening_ply`, and `rated`. This feature engineering approach prioritized interpretability while capturing the essential game dynamics that could influence outcomes.

3.3. Exploratory Data Analysis

Comprehensive exploratory analysis revealed several key patterns in the dataset:

Outcome Distribution: White achieved victory in 49.98% of games, Black won 44.91%, and draws occurred in 5.11% of matches. This distribution depicted in Figure 1, confirms the theoretical first-move advantage for White while highlighting the rarity of drawn outcomes in online play.

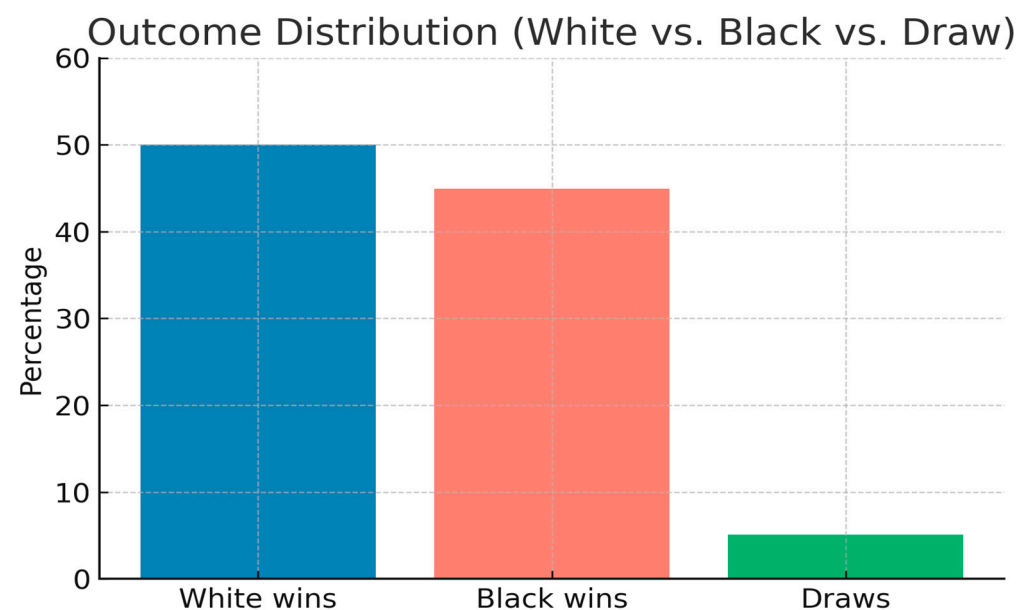


Figure 1. Games' Outcome Distribution.

Rating Distributions: As shown in Figure 2, both White and Black player ratings exhibited nearly identical distributions, clustering between 1400 and 1600 Elo points with similar variance and skewness. The mean rating difference was approximately 7.5 Elo points favoring White, indicating generally well-matched opponents.

Rating Difference (White vs. Black): The distribution of rating differences, shown in Figure 3, is nearly zero centered, indicating that the majority of games are contested between similarly rated opponents. Outliers, where rating differences exceeded 1000 points, were observed, typically representing matches between casual players and highly titled individuals. The zero-line marker in the distribution highlights perfectly even matchups.

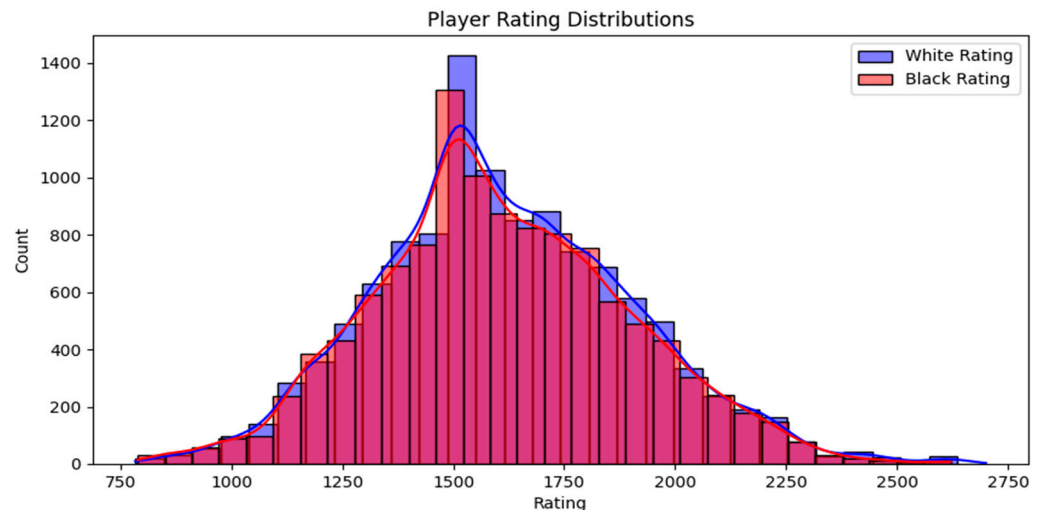


Figure 2. Player Rating Distributions.

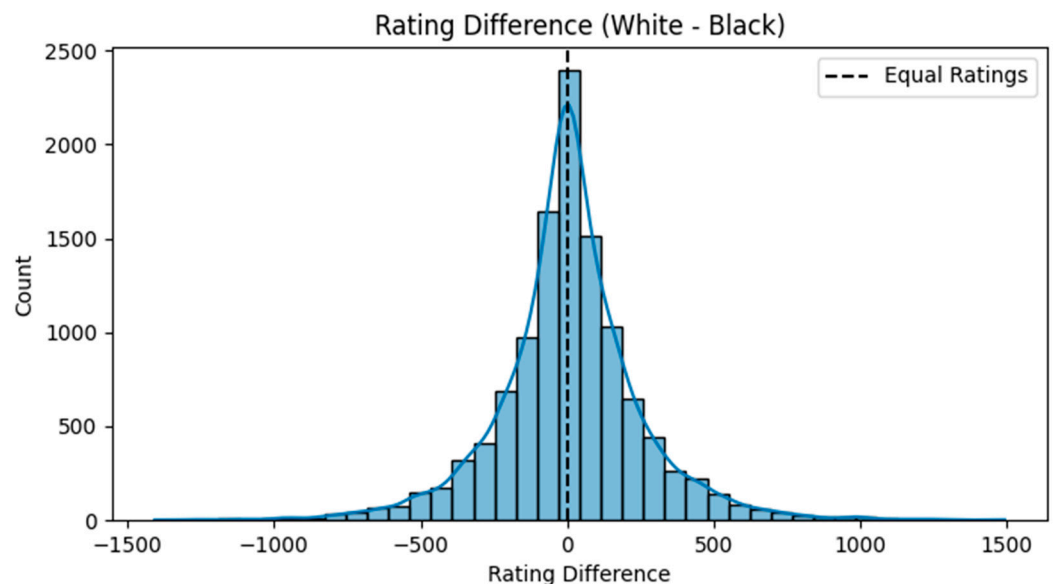


Figure 3. Rating Difference Distributions.

Rating Impact on Outcomes: Box plot analysis, shown in Figure 4, revealed clear relationships between rating differences and game outcomes. White wins correlated with positive rating differentials, Black wins with negative differentials, and draws clustered tightly around zero difference with reduced variance.

Opening Impact (Top 10 Openings): An examination of the top 10 most frequent openings in the dataset provided insights into their influence on game outcomes. A Stacked bar charts illustrated in Figure 5 shows the percentage of White wins, Black wins, and Draws for each opening revealed that certain openings, such as the Mieses–Kotrc Variation and French Defense: Knight Variation, tend to favor White. Conversely, openings like the Scandinavian Defense and Van’t Kruijs Opening showed higher Black win rates. Across all analyzed openings, draw rates remained consistently low, typically ranging from 2% to 4%.

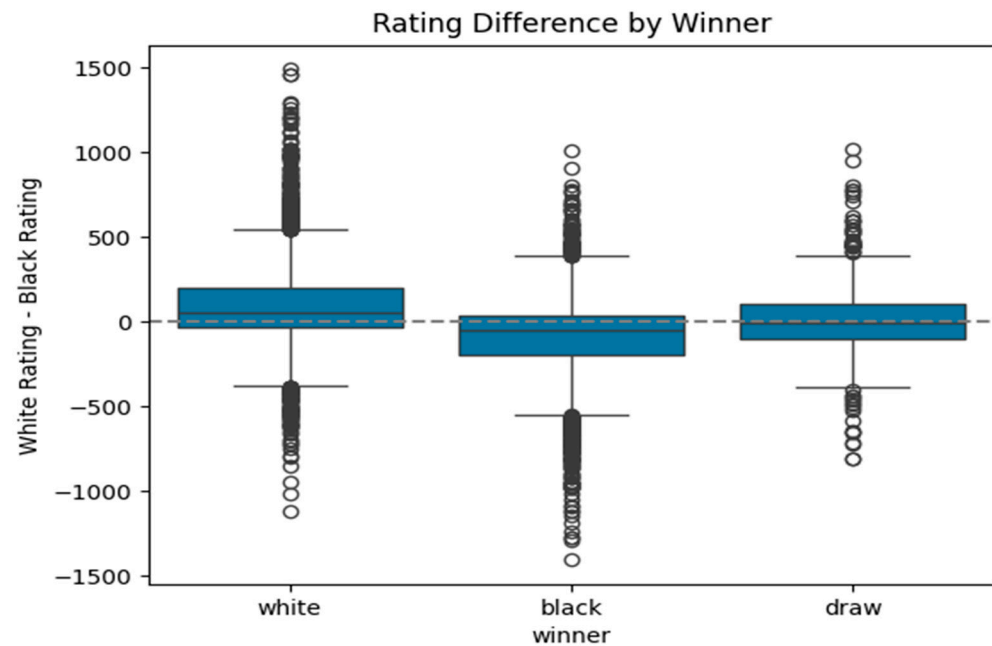


Figure 4. Box-Plots showing Rating Difference by Winner.

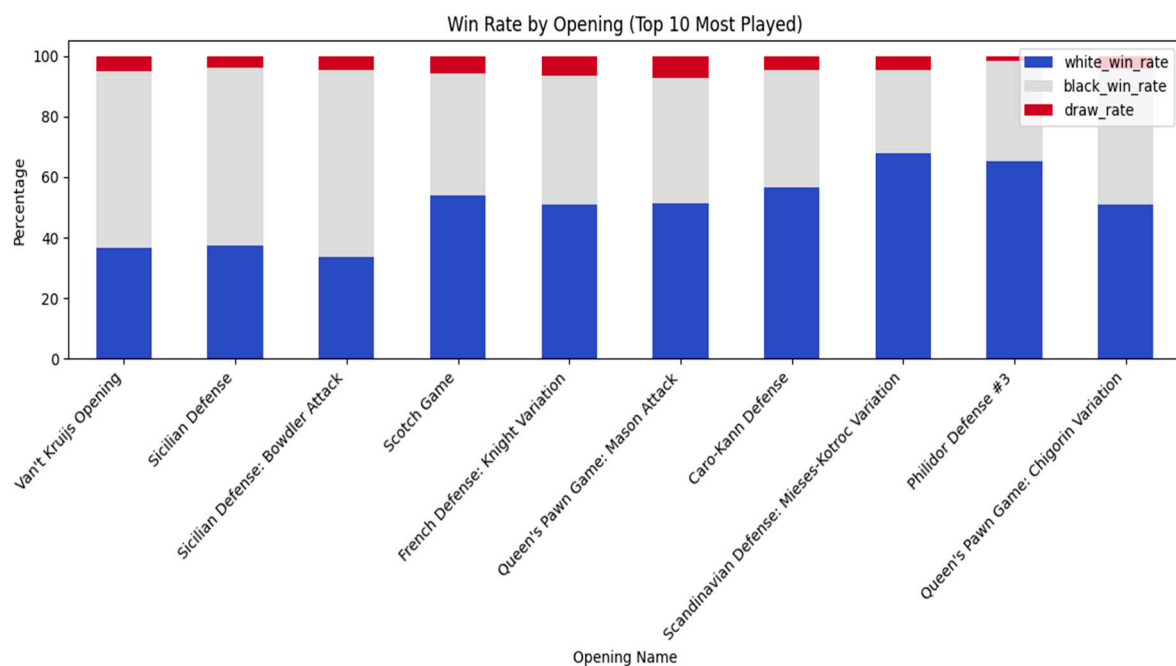


Figure 5. Examination Of the Top 10 Most Frequent Openings Moves.

4. Model Selection and Implementation

All analysis was conducted using Python (version 3.10) and the scikit-learn library. Prior to training, continuous input features were standardized using a Standard Scaler to normalize distributions, which is particularly critical for the distance-based K-Nearest Neighbors algorithm.

The processed dataset was split into a training set (80%) and a hold-out test set (20%) using stratified sampling to ensure the class distribution remained consistent across both sets. The hold-out set was completely excluded from the cross-validation process to prevent data leakage.

Four distinct machine learning algorithms were selected to represent different approaches to classification:

1. Multinomial Logistic Regression: a statistical method used to model the probability of a categorical response variable with more than two unordered outcomes as a function of a set of predictor variables. It works by estimating a set of separate binary logistic regressions, where each outcome category is compared to a single reference category [17]. The model was implemented with lbfgs solver and with One-vs-Rest (OvR) multinomial class handling.
2. Random Forest: is an ensemble machine learning algorithm that is used for both classification and regression tasks. It operates by constructing a multitude of decision trees during training and outputting the class that is the mode of the classes (for classification) or the mean prediction (for regression) of the individual trees [18]. Random Forest was used with multiple decision trees with bootstrap aggregation to reduce overfitting and improve generalization. It was implemented with the following parameters: `max_depth = 20`, `min_samples_split = 2`, `n_estimators = 100`.
3. K-Nearest Neighbors (KNN): is a non-parametric, instance-based learning algorithm that classifies a new data point based on the majority class or average of its k closest neighbors in the feature space. As a simple, yet effective, lazy learning model, its performance is highly dependent on the choice of distance metric and the value of k . [19]. Implemented with the following parameters: `knn_metric = 'manhattan'`, `knn_n_neighbors = 20`, `knn_weights = 'distance'`.
4. Histogram Gradient Boosting: is a modern, high-speed variant of the Gradient Boosting algorithm that enhances training efficiency by binning continuous feature values into discrete bins (histograms). This histogram-based approach drastically reduces the search space for optimal split points within the trees, making it computationally practical for large datasets. The best results were achieved with the following parameters: `learning_rate = 0.1`, `max_depth = 5`, `max_iter = 200`.

These four algorithms were strategically selected to represent distinct families of classifiers, allowing for a comparative analysis of how different learning paradigms handle chess metadata. Multinomial Logistic Regression serves as a linear baseline, establishing performance on linearly separable data. K-Nearest Neighbors (KNN) represents instance-based learning, testing the hypothesis that games with similar attribute vectors result in similar outcomes. Random Forest represents Bagging (Bootstrap Aggregating) ensembles, designed to reduce variance and overfitting. Finally, Histogram Gradient Boosting represents Boosting ensembles, selected for its ability to reduce bias by sequentially correcting errors of preceding predictors. This selection enables a systematic evaluation of performance progression from simple linear mappings to complex non-linear ensemble methods. While deep neural networks excel at processing complex unstructured data like board images or move sequences, ensemble methods based on decision trees often provide a superior balance of performance and interpretability for structured tabular metadata.

Each algorithm was optimized using GridSearchCV with 5-fold cross-validation to identify optimal hyperparameters. The parameter search spaces were defined based on algorithm-specific considerations and computational constraints.

A standard grid search was selected over more complex optimization methods (such as successive halving) due to the manageable size of the dataset and the efficiency of the selected hyperparameter spaces.

Evaluation Metrics

Model performance was assessed using multiple complementary metrics:

- Overall Accuracy: Proportion of correctly classified instances across all classes;
- Class-wise Precision: True positives divided by total predicted positives for each outcome class;

- Class-wise Recall: True positives divided by total actual positives for each outcome class;
- F1-Score: Harmonic mean of precision and recall for balanced performance assessment.

Feature importance analysis was conducted using algorithm-specific methods: coefficient magnitudes for logistic regression, Gini importance for Random Forest, permutation importance for KNN, and permutation importance for gradient boosting.

5. Results

5.1. Tools

Our implementation made use of the Python programming environment and several scientific libraries and frameworks:

- Programming Language: Python (version 3.10) was used to write the code for data preprocessing, model training, and evaluation.
- Machine Learning Libraries: We utilized scikit-learn for implementing Logistic Regression, K-Nearest Neighbors, Random Forest, and Histogram Gradient Boosting models.
- Data Manipulation: The pandas library was used for data loading, cleaning, and manipulation. NumPy was used for numerical computations, array handling, and feeding data into the learning algorithms efficiently.
- Data Visualization: For analysis and for creating figures, we employed Matplotlib (version 3.10.3) and Seaborn (version 0.13.2). These libraries were used to plot performance metrics (such as accuracy comparison charts) to better understand the results.

These tools and libraries provided a robust ecosystem for implementing our Chess prediction framework, from preprocessing the raw data to training complex models and visualizing their performance.

5.2. Model Performance Comparison

Table 1 and Figure 6 present comprehensive performance metrics. Accuracy is reported for both the hold-out test set and the mean of the 5-fold cross-validation. Detailed metrics (Precision, Recall, F1-Score) are reported on the hold-out set to evaluate generalization capability.

Table 1. Comparative performance metrics across all evaluated models.

Model	Hold-Out Accuracy	5-Fold CV Accuracy	Precision	Recall	F1-Score
Gradient Boosting	83.19%	83.08% $\pm$ 0.009	0.84	0.83	0.82
Random Forest	66.59%	67.35% $\pm$ 0.015	0.67	0.66	0.45
Logistic Regression	64.68%	64.73% $\pm$ 0.013	0.64	0.65	0.31
K-Nearest Neighbors	63.51%	64.60% $\pm$ 0.023	0.63	0.64	0.28

The Histogram Gradient Boosting classifier achieved superior performance across all metrics, with 83.19% hold-out accuracy and consistent cross-validation results. Notably, the close alignment between hold-out and cross-validation scores across all models indicates robust generalization without overfitting.

Error Analysis: While the Gradient Boosting model achieved high overall accuracy, an analysis of class-wise performance indicates that the majority of classification errors occurred in the 'Draw' category. Due to the high class imbalance (only 5.11% draws), simpler models like K-Nearest Neighbors and Logistic Regression struggled to distinguish draws from long decisive games, resulting in lower macro-averaged F1-scores (0.28 and

0.31, respectively) compared to Gradient Boosting (0.82), which successfully captured the specific non-linear dynamics associated with drawn games.

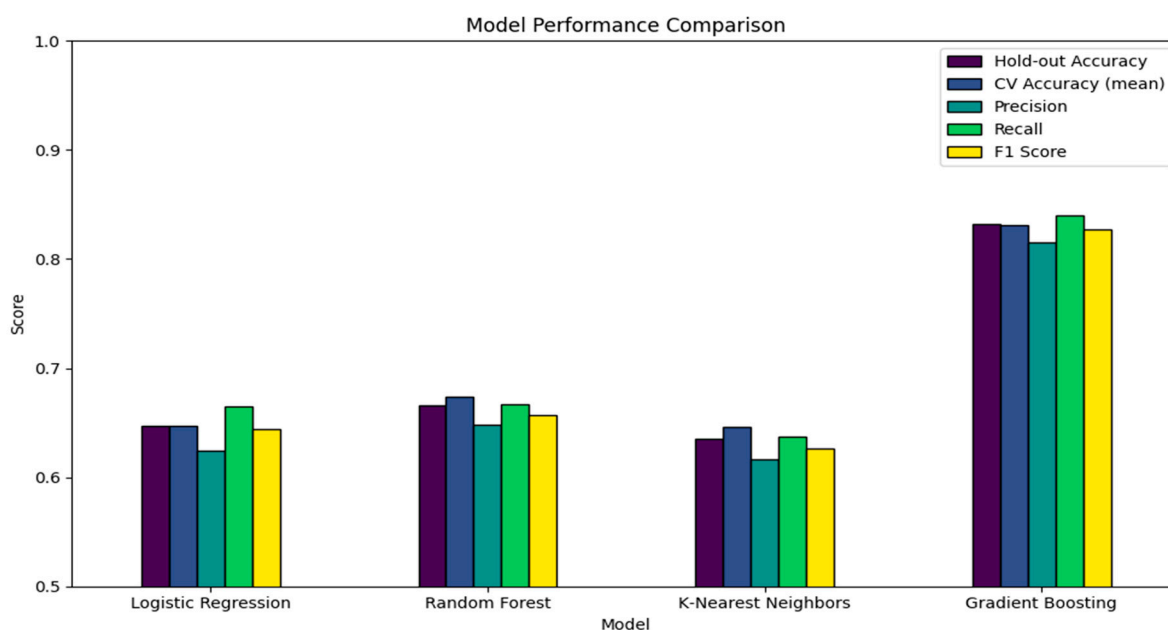


Figure 6. Comparative performance metrics across all evaluated models.

5.3. Feature Importance Analysis

Feature importance analysis values are presented in Table 2, these values reveal consistent patterns across algorithms, with some notable variations:

Table 2. Comparative feature importance values across all machine learning models.

Feature	Logistic Regression	Random Forest	KNN	Gradient Boosting
Game Length (turns)	0.20	0.22	0.25	0.35
Rating Difference (White–Black)	0.25	0.30	0.28	0.25
White Rating	0.15	0.15	0.12	0.10
Black Rating	0.15	0.15	0.12	0.10
Game Duration (s)	0.10	0.10	0.15	0.08
Opening Ply (depth)	0.08	0.05	0.04	0.04
Rated Flag	0.07	0.03	0.02	0.03
Opening Name/Type	0.00	0.00	0.02	0.05
Total (≈ 1.00)	1.00	1.00	1.00	1.00

Gradient Boosting Results: The Gradient Boosting model gives the strongest weight to game length (0.35), highlighting its power to capture non-linear patterns related to game dynamics. Rating difference (0.25) remains a major factor, but less dominant than in tree ensembles. White and black ratings (0.10 each) and game duration (0.08) have minor contributions, while opening features (0.04–0.05) and rated flag (0.03) are minimal. Gradient Boosting clearly favors dynamic, interaction-rich predictors over static metadata.

Random Forest Results: For the Random Forest model, rating difference (0.30) emerges as the dominant factor, with game length (0.22) and individual player ratings (0.15 each)

also contributing notably. Game duration (0.10) plays a secondary role, while opening ply (0.05) and rated flag (0.03) are weak predictors. Random Forest thus emphasizes skill disparity and overall game dynamics over opening characteristics

Logistic Regression Coefficients: rating difference (0.25) is the most influential predictor, closely followed by game length (0.20). Both white and black player ratings contribute moderately (0.15 each), while game duration (0.10) and opening ply (0.08) add minor effects. The rated flag (0.07) and opening name have negligible influence. Overall, the model relies mainly on rating-related variables to separate outcomes

K-Nearest Neighbors (KNN): In the KNN model, the leading determinants are rating difference (0.28) and game length (0.25), showing that distance-based classification is highly sensitive to skill gaps and game duration. White and black ratings (0.12 each) and game duration (0.15) offer moderate support, while opening features (0.04) and rated flag (0.02) contribute little. KNN depends heavily on player-level numerical features to gauge similarity between matches.

5.4. Optimal Model Configuration

The best-performing Gradient Boosting model utilized the following hyperparameters identified through grid search:

- Learning rate: 0.1;
- Maximum depth: 5;
- Maximum iterations: 200;
- Early stopping: Enabled with validation fraction of 0.1.

This configuration balanced model complexity with generalization capability, avoiding both underfitting and overfitting while maintaining computational efficiency.

6. Discussion

6.1. Key Findings and Implications

The superior performance of ensemble methods, particularly gradient boosting, demonstrates the importance of capturing non-linear interactions between game factors. The 83.19% accuracy achievement suggests that metadata contains substantial classifiable signal for chess outcomes, contradicting assumptions that chess results are primarily determined by in-game tactical decisions. While statistical methods like ANOVA could describe these relationships retrospectively, machine learning classifiers offer the distinct advantage of scalability and automation. They allow platforms to process and categorize millions of incoming game records into actionable ‘archetypes’ (e.g., ‘Upset’, ‘Grind’, ‘Stomp’) instantly, a task that manual statistical review cannot achieve at scale.

The superior performance of the Gradient Boosting classifier over the Random Forest model can be attributed to their fundamental structural differences. While Random Forest builds trees independently to reduce variance, Gradient Boosting constructs trees sequentially, where each new tree explicitly targets the residual errors of the previous ones [20]. In the context of chess outcomes, where the ‘Draw’ class represents a minority with complex decision boundaries (often overlapping with long, balanced games), the boosting approach is theoretically better suited to optimize the loss function for these hard-to-classify instances. Unlike Logistic Regression, which assumes linearity, or KNN, which struggles with high-dimensional sparsity, Gradient Boosting effectively captures the non-linear dependencies between rating disparity and game duration, resulting in the higher F1-scores observed.

The dominance of game length (turns) as a predictive feature offers several interpretations. Longer games may indicate more balanced positions where small advantages accumulate gradually, while shorter games often result from decisive tactical errors or

opening preparation advantages. This finding suggests that game tempo and complexity are fundamental determinants of outcome predictability.

Rating difference emerged as the second most important predictor, confirming the validity of Elo rating systems in capturing player skill differentials. However, the relatively modest importance of absolute player ratings indicates that skill gaps matter more than absolute skill levels for outcome determination.

6.2. Practical Applications

These findings enable several practical applications for online chess platforms focused on post-game analytics and content management:

- **Automated Content Curation:** Platforms hosting millions of games can use this classifier to automatically tag and filter completed games for user recommendations. By classifying games based on the interaction of skill gap and duration, the system can distinguish ‘High-Quality Struggles’ (Long games with specific rating dynamics) from ‘One-Sided Matches’ (Short games with high rating gaps) without manual review, improving search and discovery for users.
- **Anomaly Detection and Fair Play:** The model establishes a baseline for ‘expected’ outcomes based on ratings and game length. Completed games that deviate significantly from this classification, for example, short games where a high-rated player loses to a low-rated opponent (potential ‘sandbagging’) or long games with unusual rating disparities—can be automatically flagged as anomalies for further review by fair-play teams.
- **Player Development:** By analyzing their own completed games against the model’s classifications, players can identify patterns in their losses. For instance, if a player consistently loses games that the model classifies as “likely draws” (based on length and rating), it may indicate a specific weakness in endgame stamina or focus.

6.3. Model Interpretability and Trust

The interpretable nature of our feature set enhances model trustworthiness for practical deployment. Unlike black-box deep learning approaches, our models provide clear explanations for predictions based on readily understandable game characteristics. This transparency is crucial for user acceptance and regulatory compliance in competitive gaming environments.

7. Limitations and Future Work

7.1. Current Limitations

This study faces several limitations that restrict its scope and generalizability. The feature set included post-game attributes (game duration and turn count). While this inclusion was intentional to analyze the correlation between game complexity and outcome, it restricts the model’s utility to post hoc classification rather than pre-game forecasting. Consequently, the reported accuracy reflects the model’s ability to classify completed games based on their dynamics, rather than predict future results before they begin.

A substantial challenge in this study was the inherent class imbalance, specifically the low frequency of draws (5.11%). While synthetic data generation techniques such as SMOTE, ADASYN, or Random Duplication could theoretically improve minority class recognition, this study prioritized ecological validity by retaining the natural distribution of online games. This approach ensures that the reported accuracy reflects performance in a real-world deployment scenario where draws remain rare. However, we acknowledge that this choice creates a bias toward majority classes. Future iterations of this work aiming

purely for maximization of recall on drawn games should implement the aforementioned resampling techniques to mitigate this imbalance.

Another key limitation lies in the temporal and platform dimensions of the dataset. The analysis relied on a static snapshot of games, without accounting for evolving player behaviors, seasonal trends, or long-term rating dynamics that could influence performance over time. Furthermore, the data splitting strategy relied on stratified random sampling rather than a player-disjoint split. Given the repeated presence of frequent players in the database, this may introduce a degree of identity leakage, potentially inflating accuracy compared to a strictly player-disjoint validation scheme.

Moreover, since the dataset was drawn exclusively from Lichess, the findings may not fully generalize to other platforms with differing user bases or gameplay structures. Future work should therefore explore multi-platform datasets and temporal modeling to strengthen robustness and external validity.

7.2. Future Research Directions

Future research in chess outcome prediction could benefit from several strategic enhancements. Incorporating richer metadata such as time controls (blitz, rapid, classical), player experience metrics, opening-specific win rates, and endgame tablebase statistics would significantly strengthen predictive accuracy. These additional features would allow models to better capture the contextual and strategic nuances that influence game outcomes, especially in different phases of play.

On the modeling side, the adoption of advanced techniques such as deep learning architectures, ensemble combinations, and methods tailored for imbalanced data could yield substantial performance gains. Such approaches are particularly promising for improving the accuracy of draw predictions, which often present unique challenges due to their rarity and complex underlying dynamics. Integrating these techniques would provide a more nuanced understanding of player behavior and outcome tendencies.

Moreover, expanding the analytical scope to include temporal and cross-platform perspectives would enhance robustness and generalizability. Longitudinal analyses examining rating inflation, meta-game shifts, and seasonal variations could help models adapt to evolving player strategies over time. Validating predictive systems across different chess platforms would further reveal platform-specific influences. Finally, implementing real-time analysis systems with continuous model updating and A/B testing capabilities would bridge research and application, enabling ongoing refinement and practical deployment in live environments.

8. Conclusions

This study successfully demonstrates that machine learning approaches can effectively classify chess game outcomes using player and game metadata, with ensemble methods achieving superior performance over traditional algorithms. The Histogram Gradient Boosting classifier reached 83.19% accuracy in multiclass prediction, substantially outperforming linear and instance-based methods.

Key findings include the dominance of game length and rating difference as predictive features, the challenge of draw prediction due to class imbalance, and the effectiveness of gradient boosting in capturing non-linear feature interactions. These results provide actionable insights for chess platform optimization and player development strategies.

The interpretable nature of our models, combined with their strong predictive performance, establishes a foundation for real-world deployment in online chess environments for post-game analysis and matchmaking adjustment. While limitations exist regarding feature scope and temporal generalization, the demonstrated feasibility of outcome pre-

diction opens numerous avenues for enhancing competitive gaming experiences through data-driven approaches.

Future work should focus on expanding feature sets, addressing class imbalance through specialized techniques, and validating results across diverse chess platforms and time periods. The ultimate goal remains the development of robust, interpretable prediction systems that enhance both player experience and platform engagement in the rapidly evolving landscape of online chess.

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